Annotated Bibliography

Set Theoretic Estimation and its Mechanization

Project `thejeb`

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This bibliography tracks the sources underpinning the `thejeb` mechanization of set theoretic estimation (STE). Entries are annotated with (i) what the source establishes and (ii) its status for our Lean formalization: **seed** (a paper we are directly mechanizing), **foundational** (prior art we build on or cite), or **survey/future** (candidate for later mechanization). Every claim imported into the Lean development is cited back to a specific result here; results are trusted only once machine-checked.

Seed papers

Combettes [4] — *The Foundations of Set Theoretic Estimation*, Proc. IEEE 81(2):182–208. [**seed**] The synthesizing reference for STE. It reframes estimation away from optimizing an objective function and toward *feasibility*: a solution is acceptable iff it is consistent with all available information. Formally (§II-C), each piece of information i is a fuzzy proposition $\Psi_i : \Xi \to [0, 1]$ on the solution space Ξ; at a confidence level ψ_i it induces a *property set* $S_i = \{a \in \Xi \mid \Psi_i(a) \geq \psi_i\}$ (Eq. 3), and the *feasibility set* is $S = \bigcap_{i \in I} S_i$ (Eq. 4). A formulation is *consistent* if $S \neq \emptyset$, *fair* if the true object $h \in S$, and *ideal* if $S = \{h\}$. §III develops projection algorithms (POCS) when Ξ is a Hilbert space and the S_i are closed convex. *Mechanization status:* the crisp core (property sets, feasibility set, consistent/fair/ideal, monotonicity, invalid-information detection) is machine-checked in `Ste.Basic`; the convex layer in `Ste.Convex`. The fuzzy/graded case and the projection algorithms are future work.

Carlson [3] — *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*, Ph.D. dissertation, University of Idaho. [**seed**] Recasts STE over a *topological* solution space rather than the Hilbert space of prior work, avoiding the Optimal Bounding Ellipsoid (OBE) pathology whereby the bounding volume can spuriously expand the solution set. Applies STE to decryption: the solution space is a finite key space, each intercepted ciphertext induces a property set of keys under which it decodes to a meaningful message, and decryption succeeds when the feasible key set collapses (Shannon unicity). Named results we extracted (see `papers/notes`): Lemma 2.1 (error bound on spurious keys), Lemma 2.2 (property-set intersection no larger than the smallest set), Theorem 2.3 (property sets are AEP typical sets, tying information theory under STE), Lemma 4.1 (an S-cipher has $(|A| - |T|)!$ isomorphic keys), Theorem 4.1 (key count for a P-cipher), Theorem 5.1 (PS/SP product ciphers vs. block substitution unicity), Theorem 5.2 (a permutation cipher reduces to a substitution cipher within a symbol), Corollary 5.3 (a boundary-aligned PSP cipher is idempotent to an S-cipher; hence all block ciphers are block substitution ciphers). *Mechanization status:* the STE-decryption skeleton (cipher systems, key property sets, feasible keys, fairness of genuine traffic, unicity as ideal information, observation monotonicity) is machine-checked in `Ste.Cipher`; the counting theorems (4.1, Lemma 4.1) are the most tractable next targets, the unicity/AEP results the deepest.

Foundational

Shannon [9] — *Communication Theory of Secrecy Systems.* [**foundational**] Origin of unicity distance and key equivocation, which Carlson recasts in set theoretic terms. The set theoretic "unicity" we formalize (feasible key set = $\{k_0\}$) is the qualitative core of Shannon's quantitative unicity point.

Youla and Webb [11] — *Image Restoration by the Method of Convex Projections.* [**foundational**] POCS: iterated projection onto closed convex constraint sets converges to a feasible point. The algorithmic engine behind Combettes §III and the natural sequel to `Ste.Convex`.

Bregman [2] — relaxation / common point of convex sets. [**foundational**] The cyclic-projection method predating POCS; the ancestor of convex feasibility solvers.

Schweppe [8] — set-membership state estimation; Deller [7] — set-membership identification in DSP. [**foundational**] Early "unknown but bounded" set-membership estimation—one of the scattered traditions Combettes unifies; the source of the OBE algorithms Carlson critiques.

Survey / future mechanization

Bauschke and Borwein [1] — *On Projection Algorithms for Solving Convex Feasibility Problems*, SIAM Review. [**survey**] The definitive survey of the convex feasibility problem find $x \in \bigcap_i C_i$. Its firmly-nonexpansive-operator framework is the target for a machine-checked convergence proof of POCS atop `Ste.Convex`; Mathlib already carries the requisite inner-product-space projection theory.

Combettes [5] — *The Convex Feasibility Problem in Image Recovery.* [**survey**] Companion to the seed paper; concrete constraint sets and numerical behavior for the convex STE instantiation.

Tooling

de Moura and Ullrich [6] — Lean 4; The Mathlib Community [10] — Mathlib. [**tooling**] The proof assistant and library in which every result in this project is machine-checked. We pin Mathlib `v4.32.0` (toolchain `leanprover/lean4:v4.32.0`).

References

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[3] Albert H. Carlson. *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*. Ph.d. dissertation, University of Idaho, Moscow, Idaho, USA, May 2012. Major professors R. E. Hiromoto and R. B. Wells. Retrieved via Internet Archive snapshot (2024-07-13) of the author's ResearchGate full text, publication 349064680.

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[5] Patrick L. Combettes. The convex feasibility problem in image recovery. *Advances in Imaging and Electron Physics*, 95:155–270, 1996. doi: 10.1016/S1076-5670(08)70157-5.

[6] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, pages 625–635, 2021. doi: 10.1007/978-3-030-79876-5_37.

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[8] Fred C. Schweppe. Recursive state estimation: Unknown but bounded errors and system inputs. *IEEE Transactions on Automatic Control*, 13(1):22–28, 1968. doi: 10.1109/TAC.1968.1098790.

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